

ExamForge AI

Architecture Status Report

Phase 6 — Architecture Verification

August 2, 2026

Architecture Status Report

The ExamForge AI architecture remains unchanged from the original design. The hardening phase focused on improving the quality of the existing architecture without introducing new patterns or replacing existing components.

Technology Stack (Unchanged)

Layer	Technology	Status
Framework	Next.js 16 App Router	Production Ready
Runtime	React 19 Server Components	Production Ready
Styling	Tailwind CSS v4 + shadcn/ui	Production Ready
State	Zustand + TanStack Query	Production Ready
Auth	Supabase Auth (cookie-based)	Production Ready
Database	Supabase PostgreSQL	Production Ready
Realtime	Supabase Realtime	Production Ready
Validation	Zod 4	Production Ready
Build	Next.js Standalone Output	Production Ready

New Components Added

- `src/lib/auth/require-auth.ts` — Centralized auth helper with `requireAuth()`, `requireRole()`, `requireAnyRole()`, `getAuthUser()`, `canAccessResource()`
- `src/lib/utils/logger.ts` — Enterprise logger with structured logging, sanitization, and security-aware logging
- `.env.example` — Documented environment variables with no actual secrets

Modified Components

- `src/middleware.ts` — Complete rewrite with RBAC route mapping and role-based redirect
- `next.config.ts` — Added 7 security headers (CSP, HSTS, X-Frame-Options, etc.)
- `src/lib/types.ts` — Added 'parent' to `UserRole` type
- `src/lib/constants/routes.ts` — Added parent role to `ROLE_ROUTE_ACCESS`
- All dashboard pages — Use `requireAuth/requireAnyRole` instead of manual auth checks
- All services — Scoped queries by role and `school_id`
- All server actions — Verify auth, role, and ownership before mutations
- All API routes — Use `getAuthUser()` with proper error responses